Three Axioms of Probability
For any event $A \subset S$:

1. $P(A) \geq 0$
2. $P(S) = 1$

If $A_1, A_2, \ldots$ are pairwise mutually exclusive, then

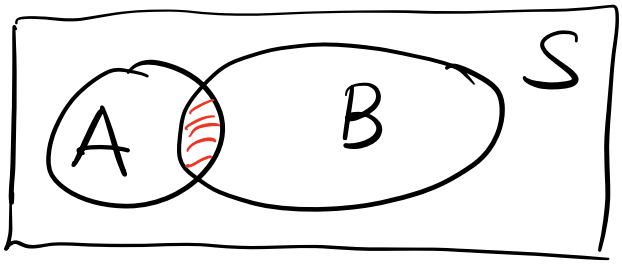
3. $P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$

Conditional Probability (continued)

Definition: Conditional Probability

The **conditional probability** of the event A given the occurrence of the event B is

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B) > 0.$$



$$\frac{P(A \cap B)}{P(B)}$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)}, \quad P(A) > 0.$$

• $B_1, B_2, \cdots, B_n$ are pairwise disjoint.
• $B_1 \cup B_2 \cup \cdots \cup B_n = S$

The Law of Total Probability

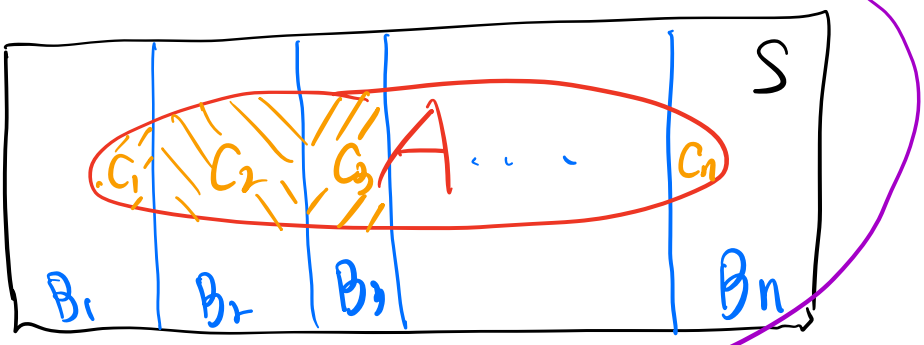
For a partition $B = \{B_1, B_2, ..., B_n\}$ and any event A in a sample space S, let

$$C_i = A \cap B_i, \qquad i = 1, 2, ..., n.$$

Then, the events C_i and C_j are mutually exclusive for $i \neq j$, and

$$A = C_1 \cup C_2 \cup ... \cup C_n.$$

$C_1, C_2, \cdots, C_n$ are pairwise disjoint.



Moreover, we have

$$P(A) = P(A \cap B_1) + P(A \cap B_2) + ...P(A \cap B_n)$$

$P(A) = P(C_1 \cup C_2 \cdots \cup C_n) \overset{\text{Axiom 3}}{=} P(C_1) + P(C_2) + \cdots + P(C_n)$

If $P(B_i) > 0$, then

$$P(A \mid B_i) = \frac{P(A \cap B_i)}{P(B_i)}$$

$\therefore P(A \cap B_i) = P(A \mid B_i) P(B_i)$

- **Law of Total Probability**

  For a partition $B = \{B_1, B_2, ..., B_n\}$ with $P(B_i) > 0$ for all i, we have

$$P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + ... + P(A|B_n)P(B_n)$$

Example (Law of Total Probability)

$S = \{1, 2, 3, 4, 5, 6\}$

Suppose a fair six-sided die is rolled once. Let

$$A = \{\text{the outcome is at least 4}\} = \{4, 5, 6\}.$$

Partition the sample space according to whether the outcome is even or odd:

$$B_1 = \{\text{the outcome is even}\} = \{2, 4, 6\},$$

$$B_2 = \{\text{the outcome is odd}\} = \{1, 3, 5\}.$$

Note: $\{B_1, B_2\}$ is a partition of the sample space S.

Since the die is fair,

$$P(B_1) = P(B_2) = \frac{3}{6} = \frac{1}{2}$$

Given that the outcome is even,

$$P(A|B_1) = \frac{P(A \cap B_1)}{P(B_1)} = \frac{P(\{4,6\})}{1/2} = \frac{2/6}{1/2} = \frac{2}{3}$$

since two outcomes in $\{2, 4, 6\}$, namely 4 and 6, are at least 4.

Given that the outcome is odd,

$$P(A|B_2) = \frac{P(A \cap B_2)}{P(B_2)} = \frac{P(\{5\})}{1/2} = \frac{1/6}{1/2} = \frac{1}{3}$$

since only 5 in $\{1, 3, 5\}$ is at least 4.

Therefore, by the law of total probability,

$$P(A) = P(A|B_1)\,P(B_1) + P(A|B_2)\,P(B_2)$$

$$= \frac{2}{3} \cdot \frac{1}{2} + \frac{1}{3} \cdot \frac{1}{2}$$

$$= \frac{1}{3} + \frac{1}{6}$$

$$= \frac{1}{2}$$

Bayes' Theorem

Bayes' theorem now lets us reverse the direction of conditioning.

Bayes' Theorem

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)}$$

$P(B|A) \overset{\text{defn.}}{=} \frac{P(B \cap A)}{P(A)}$, $P(A) > 0$ — Sub.

If $P(B) > 0$, then

$P(A|B) = \frac{P(A \cap B)}{P(B)}$

$\therefore P(A \cap B) = P(A|B)P(B)$

Example (Detecting a Transmission Error)
Suppose a communication system transmits bits over a noisy channel. Let

$$E = \{\text{a transmission error occurs}\}.$$

Suppose that

$$P(E) = 0.10.$$

An error-detection algorithm raises a warning. Let

$$W = \{\text{the detector raises a warning}\}.$$

The detector raises a warning for 90% of erroneous transmissions:

$$P(W|E) = 0.90.$$

However, it also raises a warning for 5% of correctly received transmissions:

$$P(W|E^c) = 0.05.$$

Question: If the detector raises a warning, what is the probability that a transmission error actually occurred?

Not a good way to introduce a probabity problem.

Example (Detecting a Transmission Error)

Consider a communication system that transmits one bit over a noisy channel. For each transmission, we observe whether a transmission error occurs and whether the detector raises a warning.

The sample space is

$$\Omega = \{\omega_1, \omega_2, \omega_3, \omega_4\},$$

where

Outcome	Description
ω_1	Error occurs and a warning is raised,
ω_2	Error occurs and no warning is raised,
ω_3	No error occurs and a warning is raised,
ω_4	No error occurs and no warning is raised.

Define the events

$$E = \{\omega_1, \omega_2\} = \{\text{a transmission error occurs}\},$$

and

$$W = \{\omega_1, \omega_3\} = \{\text{the detector raises a warning}\}.$$

Suppose that

$$P(E) = 0.10.$$

The detector raises a warning for 90% of erroneous transmissions:

$$P(W|E) = 0.90.$$

However, it also raises a warning for 5% of correctly received transmissions:

$$P(W|E^c) = 0.05.$$

E^c: the complement of E.

Question: If the detector raises a warning, what is the probability that a transmission error actually occurred?

W E

$$P(E|W) ?$$

$$P(E|W) = \frac{P(W|E)\,P(E)}{P(W)}$$

$$P(W) = P(W|E)\,P(E) + P(W|E^c)\,P(E^c)$$

$1 - P(E)$

$$= 0.90 \cdot 0.10 + 0.05 \cdot 0.90$$

$$= 0.135$$

$\{E, E^c\}$ is a partition of Ω.

$$\therefore P(E|W) = \frac{0.90 \cdot 0.10}{0.135}$$

$$= \frac{2}{3}$$

Independence

Definition: Two Independent Events

Events A and B are called **independent** if

$$P(A \cap B) = P(A)P(B).$$

when $P(A) \neq 0$ and $P(B) \neq 0$, we have

$P(A|B) =$

$P(B|A) =$

To be continued ... ...